Likeness subject
live consent capture

Tenant admins & creators
console · API

Connect chat / WebRTC contact

live consent capture (single-use link)

console · REST API · webhooks

End customer
browser / Connect app
+ Visavox web SDK

WebRTC

Tenant LiveKit
Cloud or self-hosted
(alternative room)

Pattern A: avatar tile in the customer's call client during self-service; human agent's Connect video after escalation.

Pattern B: avatar panel beside Connect chat or web voice (Touchpoint component).

LiveKit: avatar worker follows the LiveKit avatar contract in Visavox or tenant rooms.

Dashed line = alternative path.

Regional cell (UK · EU · US)
identical per region; tenant data never leaves its home cell

LiveKit SFU
TURN · egress
self-hosted, in-region

(or the tenant's LiveKit room)

Cognito
SSO (OIDC / SAML) · MFA

AWS WAF + API Gateway / ALB
api.{region}.visavox.ai

Amazon Bedrock
default LLM, in-region

Secrets Manager
KMS · IRSA

Services — EKS CPU nodes (Python 3.12 · FastAPI)

session service
placement · warm pool
events

control-plane API
tenants · RBAC · avatars
webhooks · audit log

consent service
liveness · face/voice
match · revocation

agent-runtime
LiveKit Agents: VAD, STT
turn, LLM, guardrails

connect-bridge
Lambda API · A2A (wss)
contact correlation

Lambda · A2A

avatar pipeline
QC · prepare · evaluate
(Step Functions)

render orchestrator
captions · C2PA signing

trust & safety
moderation · takedown
break-glass

metering
usage · GPU cost
→ Stripe export

claim GPU slot

GPU pools — EKS + Karpenter (g6 · g6e · g7e · p5 by region)

avatar-engine workers
audio → lip-synced video
N slots per GPU
Video Seal · A/V sync

TTS servers
stock + cloned voices
AudioSeal watermark

STT servers
streaming ASR
Silero VAD

avatar build jobs
Photo + Studio avatars

offline render jobs
talking-head model
+ upscale to 1080p

human agent's voice → avatar video

Data — in-region only · per-tenant KMS keys

Aurora PostgreSQL
row-level security

S3 media + artefacts
SSE-KMS per tenant

ElastiCache Redis
sessions · LiveKit

SQS + Step Functions
jobs · pipelines

AWS Backup
in-jurisdiction vaults

Observability

OpenTelemetry collector (PII scrubber)

Managed Prometheus + Grafana (SLOs)

CloudWatch Logs (IDs only)

Tenant AWS account
Amazon Connect

Connect contact
chat · WebRTC video

Flows · Visavox Lambda
A2A link (wss)

Connect AI agent / Lex
(connect_bot brain)

Human agent
custom CCP + Visavox ext.

Stripe
usage export only

C2PA trust-list CA
signing certificate

Global layer — no tenant personal or biometric data

Route 53 + CloudFront
console · web SDK

Tenant directory
DynamoDB global: slug → cell

Model licence registry
pinned hashes · licences

Public detector
watermark + C2PA check

CI/CD
GitHub Actions · Argo CD